Data Skills Better Careers!

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MODULE 4

PROBABILITY

"Better Analysts Make a Smarter Tomorrow."

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From Uncertainty to Smarter Decisions

Concepts | Formulas | Examples | Excel Practice | Real-World Applications

Topics We Will Cover

1. Introduction to Probability
2. Experiment, Trial & Outcome
3. Sample Space
4. Events
5. Types of Events
6. Basic Probability
7. Complementary Events
8. Addition Rule of Probability
9. Multiplication Rule of Probability
10. Conditional Probability
11. Independent Events
12. Dependent Events
13. Law of Total Probability
14. Bayes' Theorem
15. Applications of Probability in Business
16. Real Life Probability Examples

Build Strong Concepts | Real-World Business Use Cases | Hands-on Excel Practice | Interview Questions & Detailed Answers

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"Probability is not just a subject, It's a way to think."

Let's Connect and Grow Together!

1 Introduction to Probability

Building the Foundation for Uncertainty

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What is Probability?

Probability is the branch of mathematics that deals with uncertainty. It helps us quantify how likely an event is to occur.

> *"Probability turns uncertainty into numbers."*

Key Points

- Used to measure uncertainty
- Values range from 0 to 1 (or 0% to 100%)
- Useful in real-life decision making
- Widely used in business, finance, healthcare, and data analytics
- Foundation for many advanced topics in statistics and machine learning

Real-World Examples

- Probability of rain (weather forecasting)
- Chance of a product being defective (quality control)
- Customer purchase behaviour (marketing analytics)
- Risk of loan default (finance)
- Success rate of a treatment (healthcare)
- Predicting demand (supply chain)

Simple Example

When we roll a fair six-sided die, what is the probability of getting a 3?

$$P(3) = \frac{1}{6}$$

$$= 0.167 \text{ (or } 16.7\%)$$

There are 6 possible outcomes, and 1 favourable outcome (3), so the probability is 1 out of 6.

Terminology

Term	Meaning
Experiment	A process that produces an outcome.
Outcome	A possible result of an experiment.
Event	A set of one or more outcomes.
Sample Space (S)	The set of all possible outcomes.
Favourable Outcome	The outcomes that satisfy the event.
Probability	A number between 0 and 1 that measures the likelihood of an event.

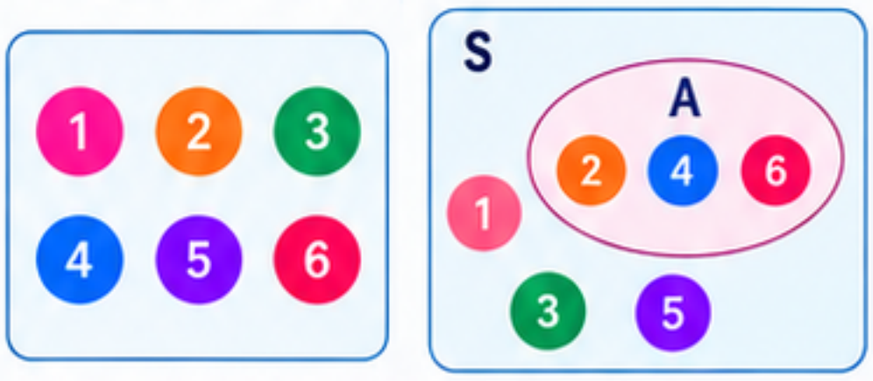
Visual Representation

Sample Space (Die Roll)

$S = \{1, 2, 3, 4, 5, 6\}$

Event A: Getting an even number

$A = \{2, 4, 6\}$



> Event A contains outcomes that satisfy the condition (even numbers).

Mathematical Notation

- $P(A)$ → Probability of event A
- S → Sample space (set of all outcomes)
- $n(S)$ → Total number of outcomes
- $n(A)$ → Number of favourable outcomes
- $0 \leq P(A) \leq 1$
- $P(A)$ can also be written in percentage form: $P(A) = 0.25 = 25\%$

Basic Probability Formula

$$P(A) = \frac{n(A)}{n(S)}$$

Where:

- $P(A)$ = Probability of event A
- $n(A)$ = Number of favourable outcomes
- $n(S)$ = Total number of possible outcomes in the sample space

This formula is used when all outcomes are equally likely.

Example – Coin Toss

A fair coin is tossed. What is the probability of getting a Head?

Sample Space: $S = \{H, T\}$

Favourable Outcomes for Head: $n(A) = 1$

Total Outcomes: $n(S) = 2$

$$P(\text{Head}) = \frac{1}{2} = 0.5 \text{ (or } 50\%)$$

Key Takeaways

- Probability helps us deal with uncertainty.
- Always define the sample space and event clearly.
- Probability values lie between 0 and 1.
- It is widely used in real-world business and analytical problems.
- This is the foundation for all upcoming probability topics.

Next Up

- Experiment, Trial & Outcome
- Sample Space
- Events
- Types of Events
- And more...

"Small concepts today, big decisions tomorrow."

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2 Experiment, Trial & Outcome

The Building Blocks of Probability

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Experiment

An experiment is a process or action that leads to a well-defined set of possible outcomes, but the exact outcome is not known in advance.

In Simple Words:
An experiment is like a game or activity that we perform to observe a result.

Example:
- Rolling a dice
- Tossing a coin
- Picking a card from a deck
- Conducting a quality check on a product

Trial

A trial is a single performance or repetition of an experiment.

In Simple Words:
A trial is one attempt of an experiment.

Example:
- Tossing a coin once → 1 trial
- Rolling a dice once → 1 trial
- Flipping a coin 5 times → 5 trials
- Selecting one student at random → 1 trial

Outcome

An outcome is a possible result of an experiment.

In Simple Words:
An outcome is what we get after performing a trial.

Example:
- Coin toss → Outcomes: Head (H), Tail (T)
- Dice roll → Outcomes: 1, 2, 3, 4, 5, 6
- Student selection → Outcomes: A, B, C, D ...
- Product test → Outcomes: Pass, Fail

Real-Life Examples

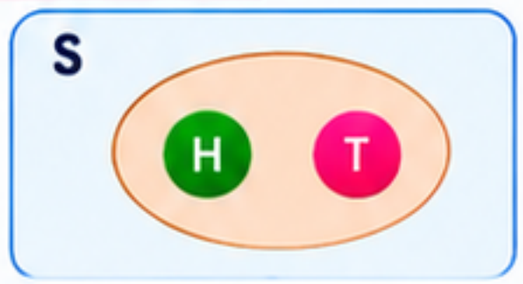
Coin Toss		**Experiment:** Toss a coin **Trial:** One toss **Outcomes:** {Head, Tail} **Number of outcomes:** 2
Dice Roll		**Experiment:** Roll a dice **Trial:** One roll **Outcomes:** {1, 2, 3, 4, 5, 6} **Number of outcomes:** 6
Card Draw		**Experiment:** Draw one card from a standard deck (52 cards) **Trial:** One draw **Outcomes:** 52 possible cards **Number of outcomes:** 52

Sample Space (S)

The sample space is the set of all possible outcomes of an experiment. It contains every outcome that can occur.

$$S = \{\text{all possible outcomes}\}$$

Examples:
- Coin Toss: $S = \{H, T\}$
- Dice Roll: $S = \{1, 2, 3, 4, 5, 6\}$
- Card Draw: $S = \{52 \text{ cards}\}$
- Select a day of the week: $S = \{Mon, Tue, Wed, Thu, Fri, Sat, Sun\}$



> The sample space is the universe of all possible outcomes.

Important Points

- An experiment has a well-defined set of outcomes.
- A trial is a single performance of the experiment.
- An outcome is a possible result of a trial.
- The sample space includes all possible outcomes.
- Outcomes are mutually exclusive (only one outcome can occur at a time).
- Probability is based on the sample space.
- Real-life problems can also be modeled using these concepts.

Types of Experiments

1. Random Experiment

- Exact outcome is not known in advance.
- Results may vary each time.
- **Example:** Coin toss, dice roll, customer arrival time.

2. Deterministic Experiment

- Outcome is known with certainty.
- No randomness involved.
- **Example:** Dropping a ball from a height (it will fall down).

Quick Check

1. Is tossing a coin an experiment, a trial or an outcome?
2. What is the sample space for a dice roll?
3. How many trials are there in rolling a dice 10 times?
4. Is "Head" an experiment, trial or outcome?
5. Can an experiment have only one possible outcome?

3 Sample Space

The Universe of All Possible Outcomes

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What is Sample Space?

The sample space (S) is the set of all possible outcomes of an experiment. It contains every outcome that can occur.

> "Sample space includes all possible results, no matter how likely or unlikely they are."

Key Points

- Contains all possible outcomes of an experiment.
- Usually denoted by S.
- Each outcome is unique.
- Outcomes are mutually exclusive.
- Used to calculate probability.
- Can be finite or infinite.

Why is it Important?

- Helps us clearly define the problem.
- Required for calculating probability.
- Helps in listing favourable outcomes.
- Used in real-world decision making.
- Foundation for advanced probability topics.

> "If you can define the sample space correctly, you're halfway to the solution."

Notation

$$S = \{\text{all possible outcomes}\}$$

- S : Sample space
- $n(S)$: Number of outcomes in the sample space
- An outcome: A single possible result

Examples

1 Coin Toss

$S = \{H, T\}$

$n(S) = 2$

2 Dice Roll

$S = \{1, 2, 3, 4, 5, 6\}$

$n(S) = 6$

3 Tossing two coins

$S = \{HH, HT, TH, TT\}$

$n(S) = 4$

Types of Sample Space

1 Finite Sample Space

- Contains a limited number of outcomes.
- Example: Coin toss, dice roll, drawing a card.
- $n(S)$ is a finite number.

2 Infinite Sample Space

- Contains unlimited outcomes.
- Example: Measuring height, time, temperature.
- $n(S)$ is infinite (cannot be counted).

More Examples

1 Drawing a Card (Standard 52 cards)

S = {all 52 cards}

$n(S) = 52$

2 Selecting a Day of the Week

S = {Mon, Tue, Wed, Thu, Fri, Sat, Sun}

$n(S) = 7$

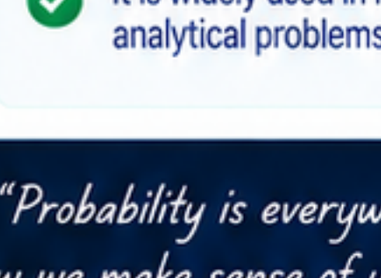
3 Product Quality Check

S = {Defective, Not Defective}

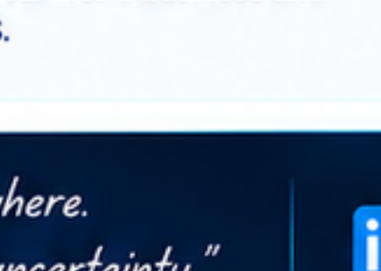
$n(S) = 2$

Visual Representation

Coin Toss



Dice Roll



Two Coins

Real-World Applications

- Risk analysis in finance (possible outcomes of market movements).
- Quality control (defective vs non-defective).
- Marketing (customer responses).
- Healthcare (treatment outcomes).
- Operations (possible demand levels).
- Data analytics (modeling possible scenarios).

> "Every real-world decision starts with understanding all possible outcomes."

Common Mistakes

- Missing some outcomes in the sample space.
- Including impossible outcomes.
- Counting the same outcome twice.
- Confusing favourable outcomes with sample space.
- Not considering order when it matters (e.g., HT ≠ TH).

Key Takeaways

- Sample space includes all possible outcomes.
- It is denoted by S.
- It can be finite or infinite.
- Defining the sample space correctly is essential for probability calculation.
- It is widely used in real-world business and analytical problems.

"Understand the possibilities to make better decisions."

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4 Events

From Outcomes to Meaningful Events

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What is an Event?

An event is a set of one or more outcomes from a sample space. It represents a specific result or a collection of results that we are interested in.

> *"An event is simply a subset of the sample space."*

Key Points

- An event is a subset of the sample space (S).
- An event can have a single outcome or multiple outcomes.
- Events are denoted by capital letters (A, B, C ...).
- An event can be simple or compound.
- Events help us calculate and understand probabilities.

> *"Small Concepts Big Opportunities"*
>
> – Tech With Shivansh

Types of Events

1. **Simple Event**
   An event with only one outcome.
2. **Compound Event**
   An event with two or more outcomes.
3. **Certain Event**
   An event that always occurs ($P = 1$).
4. **Impossible Event**
   An event that never occurs ($P = 0$).

1 Simple Event

Contains only one outcome.

Example:
Rolling a 6 on a die.

Sample Space:
$S = \{1, 2, 3, 4, 5, 6\}$

Event A: Getting a 6
$A = \{6\}$
$P(A) = \frac{1}{6}$

2 Compound Event

Contains two or more outcomes.

Example:
Getting an even number on a die.

Sample Space:
$S = \{1, 2, 3, 4, 5, 6\}$

Event B: Even number
$B = \{2, 4, 6\}$
$P(B) = \frac{3}{\cancel{6}} = \frac{1}{2}$

3 Certain Event

An event that always occurs.
It contains all outcomes in the sample space.

Example:
In a dice roll, getting a number from 1 to 6.

$S = \{1, 2, 3, 4, 5, 6\}$
Event S: $\{1, 2, 3, 4, 5, 6\}$
$P(S) = \frac{66}{66} = 1$

✓ A certain event always happens.

4 Impossible Event

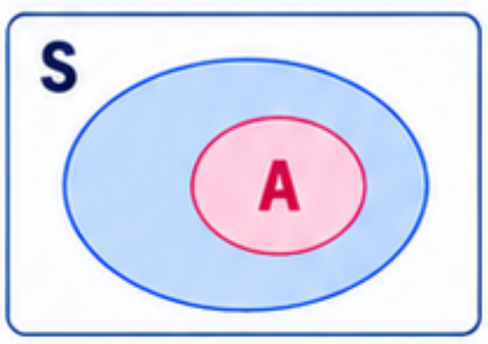
An event that never occurs.
It contains no outcomes.

Example:
Rolling a 7 on a standard die.

$S = \{1, 2, 3, 4, 5, 6\}$
Event C: $\{7\}$
$P(C) = 0$

✗ An impossible event never happens.

Event Representation (Venn Diagram)



S : Sample Space (all outcomes)

A : Event (subset of S)

> *"Every event is a part of the bigger picture called the sample space."*

Examples from Real Life

Scenario	Sample Space (S)	Event	Type
Toss a coin	{H, T}	Getting a Head {H}	Simple
Roll a die	{1, 2, 3, 4, 5, 6}	Getting an odd number {1, 3, 5}	Compound
Draw a card (from 52)	{52 cards}	Getting an Ace {A♠, A♥, A♦, A♣}	Compound
Weather forecast	{Sunny, Rainy, Cloudy}	Getting a rainy day {Rainy}	Simple

Common Mistakes

- ✗ Confusing event with outcome.
- ✗ Thinking an event can have outcomes outside the sample space.
- ✗ Assuming probability of an event without checking the sample space.
- ✗ Forgetting that an event can have multiple outcomes.
- ✗ Using lowercase letters for events.

Key Takeaways

- ✓ An event is a subset of the sample space.
- ✓ It can be simple, compound, certain or impossible.
- ✓ Events help us calculate probabilities.
- ✓ Always define the sample space first.
- ✓ Events are the building blocks for probability analysis.

5 Types of Events & Basic Probability

Understanding Events and Their Likelihood

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Types of Events

Events can be classified based on their nature, relationship and likelihood.

1. **Simple Event**
   An event with only one outcome.
   Example: Getting a 3 on a die.
2. **Compound Event**
   An event with two or more outcomes.
   Example: Getting an even number {2, 4, 6}
3. **Certain Event**
   An event that always occurs ($P = 1$).
   Example: Getting a number from 1 to 6 on a die.
4. **Impossible Event**
   An event that never occurs ($P = 0$).
   Example: Getting an 8 on a standard die.
5. **Complementary Events**
   Two events where one occurs if the other does not.
   Example: A = Getting a head, A' = Getting a tail.
6. **Mutually Exclusive Events**
   Two events that cannot occur at the same time.
   Example: Getting a 1 and a 2 on the same die roll.

Basic Probability

Probability is the measure of how likely an event is to occur. It takes values between 0 and 1.

$$P(A) = \frac{n(A)}{n(S)}$$

Where:

- $P(A)$ = Probability of event A
- $n(A)$ = Number of favourable outcomes
- $n(S)$ = Total number of outcomes in the sample space

$0 \le P(A) \le 1$

- $P(A) = 0 \rightarrow$ Impossible event
- $P(A) = 1 \rightarrow$ Certain event

Example

A fair die is rolled. What is the probability of getting a 4?

Sample Space: $S = \{1, 2, 3, 4, 5, 6\}$
Favourable Outcomes: $\{4\}$

$$P(4) = \frac{1}{6} = 0.167 \text{ (or } 16.7\%)$$

There is 1 favourable outcome out of 6 possible outcomes.

Key Takeaways

- Probability is always between 0 and 1.
- It is the ratio of favourable outcomes to total outcomes.
- A higher probability means a greater chance of occurrence.
- It is widely used in real-world decisions and risk analysis.

Complementary Events

The complement of an event A (denoted A') is the event that A does not occur.

$$P(A') = 1 - P(A)$$

Example

A coin is tossed. If $P(Head) = \frac{1}{2}$,

then $P(Tail) = 1 - \frac{1}{2} = \frac{1}{2}$

"When one event happens, its complement does not."

Mutually Exclusive Events

Two events A and B are mutually exclusive if they cannot occur at the same time.

$$P(A \cap B) = 0$$

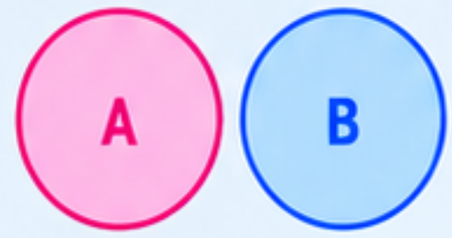
For mutually exclusive events:
$P(A \text{ or } B) = P(A) + P(B)$

Example

A die is rolled. Let A = Getting a 1, B = Getting a 2.
Since a die cannot show 1 and 2 at the same time:

$P(A \cap B) = 0$

$P(A \text{ or } B) = P(A) + P(B) = \frac{1}{6} + \frac{1}{6} = \frac{2}{6} = \frac{1}{3}$



Mutually exclusive events have no outcomes in common.

Real-World Applications

- Quality control: Probability of defective items
- Finance: Probability of market events
- Healthcare: Probability of disease occurrence
- Operations: Risk analysis in supply chain
- Marketing: Customer response probability

"Probability helps us make better decisions in an uncertain world."
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Same Concepts Real Impact!

- Data
- Analytics
- Skills
- Better Careers

6 Addition Rule & Multiplication Rule

Finding the Probability of Combined Events

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Addition Rule of Probability

The addition rule is used to find the probability that either event A or event B (or both) occurs.

General Formula:

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

Where:

- $P(A \cup B)$ = Probability of A or B (or both)
- $P(A)$ = Probability of event A
- $P(B)$ = Probability of event B
- $P(A \cap B)$ = Probability of A and B

> "We add, but subtract the overlap once (because it gets counted twice)."

For Mutually Exclusive Events

If events A and B cannot occur at the same time, then $P(A \cap B) = 0$.

$$P(A \cup B) = P(A) + P(B)$$

Example:

A card is drawn from a deck. What is the probability of getting a King or a Queen?

- $P(\text{King}) = \frac{4}{52} = \frac{1}{13}$
- $P(\text{Queen}) = \frac{4}{52} = \frac{1}{13}$
- King and Queen are mutually exclusive.

$$P(\text{King} \cup \text{Queen}) = \frac{1}{13} + \frac{1}{13} = \frac{2}{13}$$

Since one card cannot be both a King and a Queen, we simply add the probabilities.

For Non-Mutually Exclusive Events

If events A and B can occur at the same time, subtract the intersection probability.

Example:

In a survey, 40% of people like Tea (T), 30% like Coffee (C), and 10% like both. What is the probability a person likes Tea or Coffee?

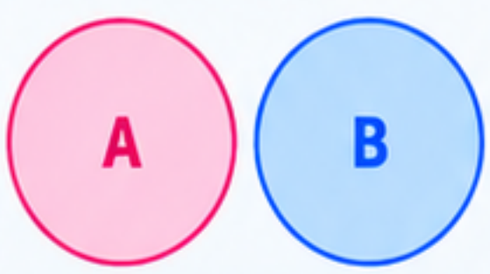
- $P(T) = 0.40$
- $P(C) = 0.30$
- $P(T \cap C) = 0.10$

$$P(T \cup C) = 0.40 + 0.30 - 0.10 = 0.60$$

We subtract the overlap because it is counted in both $P(T)$ and $P(C)$.

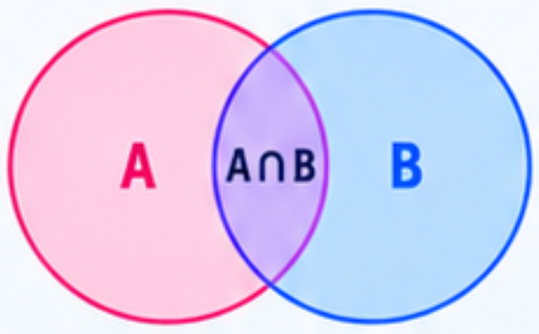
Visual Representation (Addition Rule)

Mutually Exclusive Events



$$P(A \cup B) = P(A) + P(B)$$

Non-Mutually Exclusive Events



$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

Multiplication Rule of Probability

The multiplication rule is used to find the probability that two events occur together (A and B).

Independent Events

If two events are independent (occurrence of one does not affect the other):

$$P(A \cap B) = P(A) \times P(B)$$

Example:

A coin is tossed and a die is rolled.

- $P(\text{Head}) = \frac{1}{2}$
- $P(\text{Getting a } 6) = \frac{1}{6}$

$$P(\text{Head and } 6) = \frac{1}{2} \times \frac{1}{6} = \frac{1}{12}$$

Dependent Events

If two events are dependent (occurrence of one affects the other):

$$P(A \cap B) = P(A) \times P(B \mid A)$$

Example:

A card is drawn, not replaced, and another card is drawn.

- $P(\text{First card is an Ace}) = \frac{4}{52} = \frac{1}{13}$
- $P(\text{Second card is an Ace} \mid \text{first is Ace}) = \frac{3}{51} = \frac{1}{17}$

$$P(\text{both Aces}) = \frac{1}{13} \times \frac{1}{17} = \frac{1}{221}$$

Multiplication Rule in Real Life

- Probability of multiple product purchases (e-commerce)
- Joint occurrence of risk events (finance)
- Disease diagnosis (healthcare)
- Successive defects in manufacturing (quality control)
- Customer behaviour sequences (marketing)

Common Mistakes

- Adding probabilities for 'A and B' instead of using multiplication.
- For non-mutually exclusive events, not subtracting the intersection.
- Assuming events are independent when they are actually dependent.
- Forgetting to check if events are mutually exclusive.
- Incorrectly using conditional probability.

Key Takeaways

- Use addition rule for 'A or B'.
- Subtract the intersection if events can occur together.
- Use multiplication rule for 'A and B'.
- Check if events are independent or dependent.
- Visual tools like Venn diagrams help in understanding.
- These rules are widely used in business and real-world analysis.

7 Conditional Probability

Probability with Additional Information

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What is Conditional Probability?

Conditional probability is the probability of an event occurring, given that another event has already occurred.

> *"It measures probability with additional information."*

Notation:

P(A | B) : Probability of A given that B has occurred.

Formula

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}$$

Where:

- $P(A \mid B)$ = Probability of A given B
- $P(A \cap B)$ = Probability of A and B
- $P(B)$ = Probability of B
- $P(B) > 0$

We only use conditional probability when we have information about the occurrence of another event.

Example 1 – Card Draw

A card is drawn from a standard deck. Given that the card is a King, what is the probability that it is a red card?

Solution:

- Event A: Card is red
- Event B: Card is a King
- $P(A \cap B) = \frac{2}{52}$ (King of Hearts or King of Diamonds)
- $P(B) = \frac{4}{52}$ (Total Kings)

$$P(A \mid B) = \frac{\left(\frac{2}{52}\right)}{\left(\frac{4}{52}\right)} = \frac{1}{2} = 0.5$$

> Given that the card is a King, the probability that it is red is **50%**.

Example 2 – Medical Test

A disease affects 1% of a population. A test correctly identifies 95% of diseased people (sensitivity) and 90% of healthy people (specificity). If a person tests positive, what is the probability that they actually have the disease?

Given:

- $P(D) = 0.01$
- $P(+ \mid D) = 0.95$
- $P(+ \mid D') = 0.10$

Solution:

$$P(D \mid +) = \frac{0.95 \times 0.01}{0.95 \times 0.01 + 0.10 \times 0.99}$$

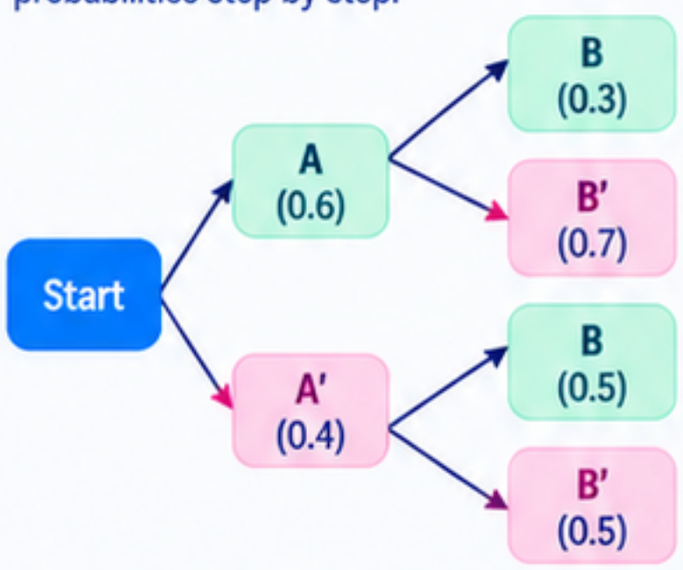
$$= \frac{0.0095}{0.0095 + 0.099}$$

$$= 0.0872 \ (8.72\%)$$

> Even if the test is positive, the probability of actually having the disease is only **8.72%**.

Tree Diagram

Tree diagrams help us visualize conditional probabilities step by step.



Example: Find **P(B)**

$$P(B) = P(A)P(B \mid A) + P(A')P(B \mid A')$$

$$= (0.6 \times 0.3) + (0.4 \times 0.5)$$

$$= 0.18 + 0.20 = 0.38$$

Real-World Applications

- **Healthcare** — Disease probability, test accuracy
- **Finance** — Credit risk, fraud detection
- **Marketing** — Customer behavior, conversion probability
- **Operations** — Failure analysis, quality control
- **HR** — Hiring probability, performance prediction
- **Insurance** — Claim prediction, risk assessment

> Conditional probability helps us make smarter decisions in uncertain situations."

Bayes' Theorem (Intro)

Bayes' Theorem helps us reverse the conditional probability.

$$P(A \mid B) = \frac{P(B \mid A)P(A)}{P(B)}$$

Where:

- P(A | B) = Posterior probability (after seeing evidence)
- P(B | A) = Likelihood (probability of evidence given A)
- P(A) = Prior probability (initial probability)
- P(B) = Total probability of evidence

Key Takeaways

- Conditional probability uses additional information.
- $P(A \mid B) = \frac{P(A \cap B)}{P(B)}$.
- Tree diagrams help visualize steps.
- Bayes' Theorem is useful for real-world predictions.
- Widely used in business, healthcare, and analytics.

Common Mistakes

- Using $P(A)/P(B)$ instead of $P(A \cap B)/P(B)$.
- Forgetting the condition (given B).
- Using conditional probability when $P(B) = 0$.
- Confusing $P(A \mid B)$ with $P(B \mid A)$.
- Not checking if events are independent.
- Ignoring real-world context in interpretation.

"More information leads to better probabilities and better decisions."
– Tech With Shivansh

8 Independent Events & Dependent Events

When Events Affect Each Other (or Not)

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What are Independent Events?

Two events are independent if the occurrence of one event does not affect the probability of the other event.

> *"The outcome of one event does not change the chances of the other."*

Mathematical Condition:

$$P(A \cap B) = P(A) \times P(B)$$

or equivalently,

$$P(A \mid B) = P(A) \quad \text{and} \quad P(B \mid A) = P(B)$$

Example 1 – Coin Toss

A coin is tossed twice. Let event A be 'first toss is Head' and event B be 'second toss is Head'. Are A and B independent?

Solution:

- $P(A) = 1/2$
- $P(B) = 1/2$
- $P(A \cap B) = 1/4$
- $P(A) \times P(B) = 1/2 \times 1/2 = 1/4$

Since $P(A \cap B) = P(A) \times P(B)$, events A and B are independent.

Example 2 – Dice Roll

A die is rolled. Let event A be 'getting a 3' and event B be 'getting an even number'. Are A and B independent?

Solution:

- $P(A) = 1/6$
- $P(B) = 3/6 = 1/2$
- $P(A \cap B) = 0$ (3 is not an even number)
- $P(A) \times P(B) = 1/6 \times 1/2 = 1/12$

Since $P(A \cap B) \neq P(A) \times P(B)$, events A and B are independent.

What are Dependent Events?

Two events are dependent if the occurrence of one event affects the probability of the other event.

> *"The outcome of one event changes the chances of the other."*

Mathematical Condition:

$$P(A \mid B) \neq P(A) \quad \text{or} \quad P(B \mid A) \neq P(B)$$

Joint Probability Formula:

$$P(A \cap B) = P(A) \times P(B \mid A)$$

or $P(A \cap B) = P(B) \times P(A \mid B)$

Example 1 – Cards without Replacement

A card is drawn from a deck, not replaced, and then another card is drawn. Let A be 'first card is an Ace' and B be 'second card is an Ace'. Are A and B independent?

Solution:

- $P(A) = 4/52 = 1/13$
- $P(B \mid A) = 3/51 = 1/17$
- $P(A \cap B) = 1/13 \times 1/17 = 1/221$
- $P(A) \times P(B) = 1/13 \times 1/13 = 1/169$

Since $P(A \cap B) \neq P(A) \times P(B)$, events A and B are dependent.

Example 2 – Drawing Balls

A bag has 3 red balls and 2 blue balls. Two balls are drawn without replacement. Let A be 'first ball is red' and B be 'second ball is red'. Are A and B independent?

Solution:

- $P(A) = 3/5$
- $P(B \mid A) = 2/4 = 1/2$
- $P(A \cap B) = 3/5 \times 1/2 = 3/10$
- $P(A) \times P(B) = 3/5 \times 3/5 = 9/25$

Since $P(A \cap B) \neq P(A) \times P(B)$, events A and B are dependent.

Independent vs Dependent Events

Basis	Independent Events	Dependent Events
Definition	One event does not affect the other.	One event affects the other.
Formula	$P(A \cap B) = P(A) \times P(B)$	$P(A \cap B) = P(A) \times P(B \mid A)$
Conditional Probability	$P(A \mid B) = P(A)$	$P(A \mid B) \neq P(A)$
Example	Coin toss, dice roll, lottery draw (separate).	Cards without replacement, drawing balls without replacement.
Real-World	Random system failures, surveys on different people.	Customer repeat purchase, medical tests, quality checks.

Key Takeaways

- Independent events do not affect each other.
- Dependent events affect each other.
- Use multiplication rule for dependent events.
- Check with conditional probabilities.
- Real-world problems are often dependent.
- Always identify if events are independent or dependent first.

Common Mistakes

- Assuming all events are independent.
- Using $P(A) \times P(B)$ for dependent events.
- Not checking if replacement is involved.
- Confusing conditional probability with joint probability.
- Forgetting to interpret results in context.

> *"Understanding dependence helps us think more clearly about real-world uncertainties."*
> – Tech With Shivansh

9 Law of Total Probability

Breaking a Complex Problem into Simpler Parts

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What is the Law of Total Probability?

The law of total probability is used to find the probability of an event by considering all possible ways (mutually exclusive and exhaustive events) in which that event can occur.

> *"It helps us find the overall probability by combining conditional probabilities from different groups."*

When to Use?

- When the sample space can be divided into mutually exclusive and exhaustive events.
- When we know the conditional probabilities of an event under each group.
- Commonly used in real-life business and analytical problems.

Formula

If $B_1, B_2, \ldots, B_n$ are mutually exclusive and exhaustive events, then for any event A:

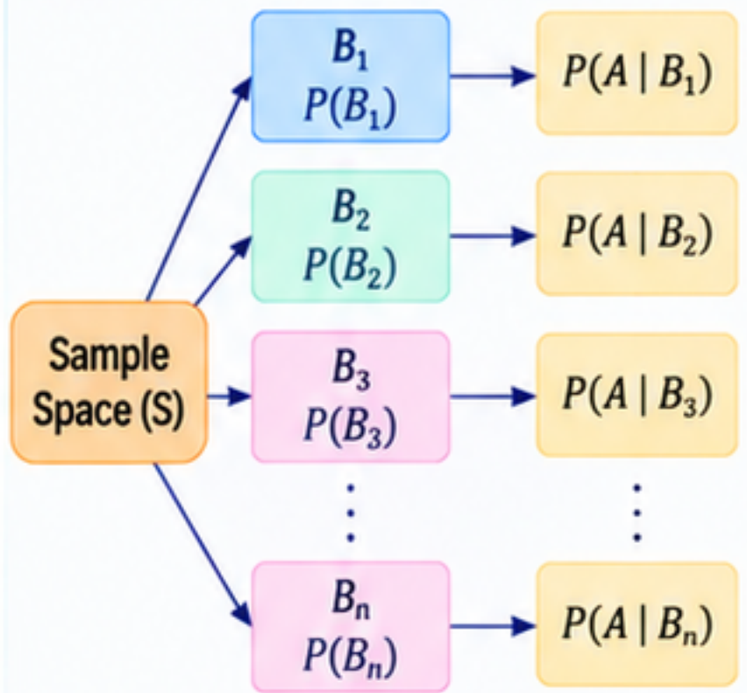
$$P(A) = \sum_{i=1}^{n} P(A \mid B_i)P(B_i)$$

Where:

- B_i are mutually exclusive and exhaustive events (i.e., they cover the entire sample space).
- $P(A \mid B_i)$ is the probability of event A occurring given B_i.
- $P(B_i)$ is the probability of event B_i.

The events $B_1, B_2, \ldots, B_n$ must be:
1. Mutually exclusive (cannot occur together)
2. Exhaustive (cover all possible outcomes)

Visual Representation



> *"We break the problem into smaller cases, find the probability in each case, and then combine them."*

Example 1 – Quality Control

A company has three manufacturing plants:
- Plant 1 produces 50% of the items, with 2% defective.
- Plant 2 produces 30% of the items, with 3% defective.
- Plant 3 produces 20% of the items, with 5% defective.

What is the probability that a randomly selected item is defective?

Solution:

Let D = Item is defective
B_1 = Plant 1, B_2 = Plant 2, B_3 = Plant 3

$$P(D) = P(D \mid B_1)P(B_1) + P(D \mid B_2)P(B_2) + P(D \mid B_3)P(B_3)$$

$$= (0.02 \times 0.50) + (0.03 \times 0.30) + (0.05 \times 0.20)$$

$$= 0.01 + 0.009 + 0.01 = 0.029$$

The probability that an item is defective is **2.9%**.

Example 2 – Medical Diagnosis

A disease occurs in 1% of the population. A test is 99% accurate for diseased people and 95% accurate for non-diseased people. What is the probability that a randomly selected person tests positive?

Solution:

Let T = Test is positive
B_1 = Has disease, B_2 = Does not have disease

- $P(B_1) = 0.01$
- $P(B_2) = 0.99$
- $P(T \mid B_1) = 0.99$
- $P(T \mid B_2) = 0.05$

$$P(T) = P(T \mid B_1)P(B_1) + P(T \mid B_2)P(B_2)$$

$$= (0.99 \times 0.01) + (0.05 \times 0.99)$$

$$= 0.0099 + 0.0495 = 0.0594$$

The probability that a person tests positive is **5.94%**.

Real-World Applications

- **Manufacturing** — Product defect probability
- **Healthcare** — Disease testing and diagnosis
- **Marketing** — Customer response across segments
- **Finance** — Credit default probability
- **Human Resources** — Employee attrition analysis
- **Risk Management** — Combined risk from multiple sources

> "The law of total probability is widely used in business, analytics and decision making."

Key Takeaways

- Used when the sample space is divided into mutually exclusive and exhaustive events.
- Find the probability of an event by combining conditional probabilities.
- Applicable in many real-world scenarios.
- Accuracy depends on correct identification of the groups (B_i).

Common Mistakes

- Events are not mutually exclusive.
- Events are not exhaustive (don't cover the entire sample space).
- Incorrect conditional probabilities.
- Forgetting to multiply by $P(B_i)$.
- Using the formula when groups overlap.
- Calculation errors in summation.

Remember

- Divide → Find Conditional → Multiply → Add
- Always check mutual exclusivity and exhaustiveness.
- Useful for complex real-world problems.

> *"Complex problems become simple when we break them into cases."*
> – Tech With Shivansh

10 Bayes' Theorem

Updating Probabilities with New Information

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What is Bayes' Theorem?

Bayes' Theorem is used to update our belief about an event after getting new evidence or information.

> *"It helps us revise probabilities when we get new information."*

Formula

$$P(A \mid B) = \frac{P(B \mid A)P(A)}{P(B)}$$

Where:

- $P(A \mid B)$ = Posterior probability (updated probability)
- $P(B \mid A)$ = Likelihood (probability of evidence given the event)
- $P(A)$ = Prior probability (initial belief)
- $P(B)$ = Marginal probability (total probability of evidence)

Understanding the Terms

Prior Probability ($P(A)$)

- Initial belief before seeing the evidence.
- Based on past data or experience.
- **Example:** Probability of having a disease in the population.

Likelihood ($P(B \mid A)$)

- Probability of observing the evidence given the event is true.
- **Example:** Probability of a test being positive given a person has the disease.

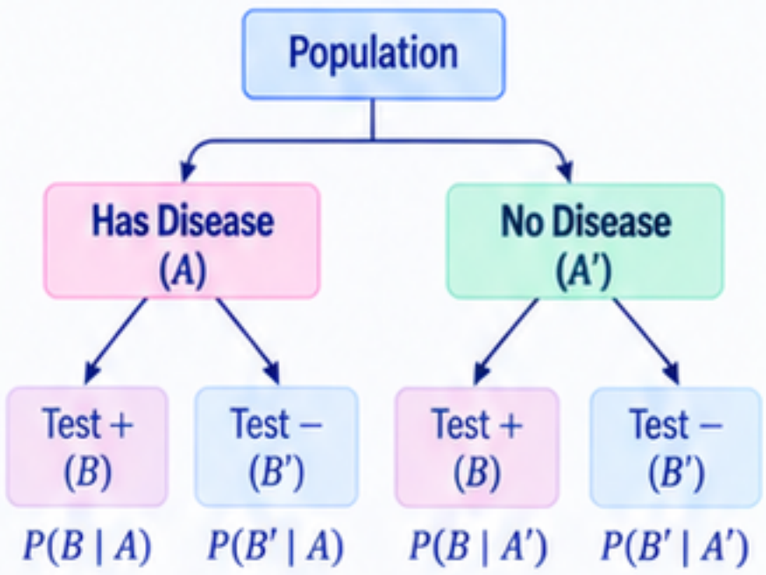
Marginal Probability ($P(B)$)

- Total probability of observing the evidence (over all possible cases).
- Calculated using the law of total probability.

Posterior Probability ($P(A \mid B)$)

- Updated probability after seeing the evidence.
- This is what we want to find.

Visual Representation



Total Probability of Evidence ($P(B)$)

$$P(B) = P(B \mid A)P(A) + P(B \mid A')P(A')$$

> *Bayes' Theorem flips the probability from $P(B \mid A)$ to $P(A \mid B)$."*

Example 1 – Medical Testing

A disease affects 1% of the population. A test for the disease is 99% accurate for people who have the disease (sensitivity) and 95% accurate for people who do not have the disease (specificity). If a person tests positive, what is the probability that they actually have the disease?

Given:

- $P(A) = 0.01$ (1% have the disease)
- $P(B \mid A) = 0.99$ (true positive)
- $P(B \mid A') = 0.05$ (false positive rate, since specificity is 95%)

Solution:

$$P(B) = (0.99 \times 0.01) + (0.05 \times 0.99)$$
$$= 0.0099 + 0.0495 = 0.0594$$

$$P(A \mid B) = \frac{0.99 \times 0.01}{0.0594} = 0.1667 \ (16.67\%)$$

Even with a highly accurate test, the probability of actually having the disease is only **16.67%**!

Example 2 – Spam Filtering

In an email system, 20% of all emails are spam. The filter correctly identifies 95% of spam emails and incorrectly marks 3% of non-spam emails as spam. If an email is marked as spam, what is the probability that it is actually spam?

Given:

- $P(A) = 0.20$ (email is spam)
- $P(B \mid A) = 0.95$ (correctly identified spam)
- $P(B \mid A') = 0.03$ (non-spam marked as spam)

Solution:

$$P(B) = (0.95 \times 0.20) + (0.03 \times 0.80)$$
$$= 0.19 + 0.024 = 0.214$$

$$P(A \mid B) = \frac{0.95 \times 0.20}{0.214} = 0.8879 \ (88.79\%)$$

If an email is marked as spam, there is an **88.79%** chance that it is actually spam.

Example 3 – Quality Control

In a factory, 5% of items are defective. A test correctly identifies 90% of defective items and incorrectly flags 4% of good items as defective. If an item is flagged as defective, what is the probability that it is actually defective?

Given:

- $P(A) = 0.05$ (item is defective)
- $P(B \mid A) = 0.90$ (test detects defective)
- $P(B \mid A') = 0.04$ (false alarm)

Solution:

$$P(B) = (0.90 \times 0.05) + (0.04 \times 0.95)$$
$$= 0.045 + 0.038 = 0.083$$

$$P(A \mid B) = \frac{0.90 \times 0.05}{0.083} = 0.5422 \ (54.22\%)$$

If an item is flagged as defective, there is a **54.22%** chance that it is actually defective.

Key Takeaways

- Bayes' Theorem updates probabilities using new evidence.
- It combines prior probability and likelihood.
- Useful in real-world scenarios like healthcare, spam filtering, fraud detection, and quality control.
- Always calculate $P(B)$ using the law of total probability.
- Helps in making better decisions under uncertainty.

Common Mistakes

- Using $P(B \mid A)$ instead of $P(A \mid B)$.
- Forgetting to calculate $P(B)$ correctly.
- Confusing prior and posterior probabilities.
- Ignoring all possible cases in $P(B)$.
- Misinterpreting the result in real-world context.

Real-World Applications

- Medical diagnosis and test evaluation
- Spam filtering in email systems
- Fraud detection in finance
- Quality control in manufacturing
- Market research and customer behavior
- Risk assessment and decision making

11 Applications of Probability in Business

From Theory to Real-World Decisions

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Why Probability in Business?

Businesses operate in uncertain environments. Probability helps in:

- Estimating risk and opportunities
- Making data-driven decisions
- Forecasting demand and sales
- Managing uncertainty
- Improving operational efficiency

> *"Probability turns uncertainty into a manageable risk."*
> – Tech With Shivansh

1. Demand Forecasting

Problem:
A product has a 70% chance of being in high demand and 30% chance of low demand.

Question:
What is the expected demand if high demand is 1,000 units and low demand is 400 units?

Solution:

- $P(\text{High}) = 0.70$ • $P(\text{Low}) = 0.30$
- $\text{Expected Demand} = (0.70 \times 1{,}000) + (0.30 \times 400)$
  $= 700 + 120 = 820 \text{ units}$

The company should plan for an expected demand of **820** units.

2. Risk Analysis

Problem:
A company faces a 5% chance of machine breakdown in a month. If it happens, the loss is ₹10,00,000. What is the expected loss?

Solution:

- $P(\text{Breakdown}) = 0.05$
- $\text{Loss} = ₹10{,}00{,}000$
- $\text{Expected Loss} = 0.05 \times 10{,}00{,}000$
  $= ₹50{,}000$

On average, the company can expect ₹50,000 loss per month due to machine breakdown risk.

3. Decision Making

Problem:
A company is launching a new product. There is a 60% chance of success with a profit of ₹5,00,000. and a 40% chance of failure with a loss of ₹2,00,000. Should the company launch the product?

Solution:

- $P(\text{Success}) = 0.60$, Profit $= ₹5{,}00{,}000$
- $P(\text{Failure}) = 0.40$, Loss $= ₹2{,}00{,}000$
- $\text{Expected Value} = (0.60 \times 5{,}00{,}000) + (0.40 \times (-2{,}00{,}000))$
  $= 3{,}00{,}000 - 80{,}000 = ₹2{,}20{,}000$

Since the expected value is positive, the company should launch the product.

4. Customer Segmentation

Problem:
From past data, 20% of customers are high-value (H) and 80% are regular (R). A purchase is made by 70% of high-value customers and 30% of regular customers. What is the probability that a randomly selected customer makes a purchase?

Solution (Law of Total Probability):

- $P(H) = 0.20$ • $P(R) = 0.80$
- $P(\text{Purchase} \mid H) = 0.70$
- $P(\text{Purchase} \mid R) = 0.30$

$$P(\text{Purchase}) = 0.20 \times 0.70 + 0.80 \times 0.30 = 0.14 + 0.24 = 0.38$$

There is a 38% chance that a randomly selected customer makes a purchase.

5. Quality Control

Problem:
A factory produces 2% defective items. What is the probability of getting at least one defective item in a sample of 10 items?

Solution:

- $P(\text{Defective}) = 0.02$, $P(\text{Non-defective}) = 0.98$
- $P(\text{No defective in 10}) = 0.98^{10} = 0.8171$
- $P(\text{At least one defective}) = 1 - 0.8171 = 0.1829$

There is an **18.29%** chance of getting at least one defective item in a sample of 10.

6. Marketing Campaign

Problem:
An email campaign has a 25% chance of being opened. What is the probability that at least one out of 5 customers opens the email?

Solution:

- $P(\text{Open}) = 0.25$, $P(\text{Not Open}) = 0.75$
- $P(\text{No one opens}) = 0.75^5 = 0.2373$
- $P(\text{At least one opens}) = 1 - 0.2373 = 0.7627$

There is a **76.27%** chance that at least one customer opens the email.

Other Business Use Cases

- Fraud detection (probability of fraud)
- Inventory management (stock-out risk)
- Customer churn prediction
- Credit risk assessment (loan defaults)
- A/B testing in marketing
- Project completion probability
- Insurance claim modeling
- Pricing and revenue optimization

> *"Probability is everywhere in business – you just need to learn how to see it."*
> – Tech With Shivansh

Key Takeaways

- Probability helps in real-world decision making.
- Used in demand forecasting, risk analysis, marketing, quality control, and more.
- Expected value helps evaluate opportunities.
- Combines well with conditional probability and Bayes' theorem.
- Widely used in modern business analytics and data science roles.
- Always interpret the result in business context.

Data Creates Opportunites!

Common Mistakes

- Ignoring real-world context.
- Using wrong probabilities.
- Not considering all possible outcomes.
- Misinterpreting expected value.

12 Real-Life Probability Examples

Case Studies, Interview Questions & Answers

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1 Retail – Product Returns

A retail company knows that 5% of the products are defective. What is the probability that out of 3 randomly selected products, at least one is defective?

Solution:

- $P(\text{Defective}) = 0.05$
- $P(\text{No defective}) = 0.95$
- $P(\text{At least one defective}) = 1 - P(\text{No defective})$
  $= 1 - (0.95)^3$
  $= 1 - 0.8574 = 0.1426\ (14.26\%)$

There is a **14.26%** chance that at least one product is defective.

2 Marketing – Campaign Success

An email campaign has a 20% chance of getting a click. If 5,000 emails are sent, what is the probability that at least 900 people click?

Let $X \sim \text{Binomial}(n = 5000, p = 0.20)$

Mean $(\mu) = n \times p = 1000$

Standard Deviation $(\sigma) = \sqrt{n \times p \times (1-p)}$

$= \sqrt{5000 \times 0.2 \times 0.8} = 28.28$

Using Normal Approximation:

$$Z = \frac{900 - 1000}{28.28} = -3.54$$

$$P(X \geq 900) = P(Z \geq -3.54) = 0.9998$$

There is a **99.98%** chance that at least 900 people click the email.

3 Finance – Credit Risk

Past data shows 2% of loan applicants default. If 10 applicants are selected, what is the probability that exactly 2 will default?

Solution:

- $n = 10,\ p = 0.02,\ k = 2$
- $P(X = 2) = {}^{10}C_2\,(0.02)^2\,(0.98)^8$
  $= 45 \times 0.0004 \times 0.8507$
  $= 0.0153\ (1.53\%)$

There is a **1.53%** chance that exactly 2 applicants will default.

4 Operations – Machine Downtime

A machine has a 3% chance of breaking down on any given day. What is the probability that it works for 30 consecutive days without a breakdown?

Solution:

- $P(\text{Works}) = 1 - 0.03 = 0.97$
- $P(\text{Works for 30 days}) = (0.97)^{30}$
  $= 0.401\ (40.1\%)$

There is a **40.1%** chance that the machine will work for 30 days without a breakdown.

5 HR – Employee Attrition

The probability that an employee leaves in a year is 10%. What is the probability that at least one employee leaves in a team of 8?

Solution:

- $P(\text{Leaves}) = 0.10$
- $P(\text{At least one leaves}) = 1 - (0.90)^8$
  $= 1 - 0.4305$
  $= 0.5695\ (56.95\%)$

There is a **56.95%** chance that at least one employee will leave.

6 Healthcare – Disease Testing

A test has 98% sensitivity and 95% specificity. If 1% of the population has the disease and a person tests positive, what is the probability that they actually have the disease?

Solution (Bayes' Theorem):

- $P(D) = 0.01$
- $P(+ \mid D) = 0.98$
- $P(+ \mid D') = 0.05$

$$P(D \mid +) = \frac{0.98 \times 0.01}{0.98 \times 0.01 + 0.05 \times 0.99}$$
$$= 0.165\ (16.5\%)$$

Even with a positive test, there is only a **16.5%** chance of actually having the disease.

8 Interview Questions & Answers

Q1. What is the difference between independent and dependent events?
A: Independent events do not affect each other's probability. Dependent events affect each other's probability.

Q2. When do we use the law of total probability?
A: When an event can occur through multiple mutually exclusive and exhaustive events.

Q3. What is Bayes' theorem used for?
A: To update probabilities based on new evidence.

Q4. Give a real-life example of conditional probability.
A: Probability of having a disease given a positive test result.

Q5. How is probability used in business?
A: In demand forecasting, risk analysis, quality control, marketing, and decision making.

Q6. If P(A) = 0.3 and P(B | A) = 0.5, what is P(A ∩ B)?
A: $P(A \cap B) = P(A) \times P(B \mid A) = 0.3 \times 0.5 = 0.15$

Key Takeaways from This Module

- Probability helps us model uncertainty.
- Independent and dependent events are used in real-world problems.
- Law of total probability helps break complex problems.
- Bayes' theorem updates probabilities using new evidence.
- Probability is widely used in business, healthcare, finance, marketing and operations.
- Real-life applications make probability more practical and valuable.
- These concepts are frequently asked in interviews.

"Probability is not just a theory, it's a powerful tool to solve real-world problems."
– Tech With Shivansh

Common Interview Tips

- Understand the concept, not just the formula.
- Explain with real-world examples.
- Show step-by-step calculations.
- Be clear with assumptions.
- Practice explaining in simple words.
- Solve past interview questions.
- Connect the answer to business impact.
- Stay confident and structured.

Topics Covered in This Module

1. What is Probability?
2. Sample Space & Events
3. Rules of Probability
4. Conditional Probability
5. Independent & Dependent Events
6. Law of Total Probability
7. Bayes' Theorem
8. Applications in Business
9. Real-Life Examples
10. Interview Questions & Answers